

v4.2 — Hotspot Triage (10×, Path Foundation)

E2E outcome: after QC, the system overlays a tumor-probability heatmap and proposes editable hotspot ROIs; the pathologist adjusts polygons and confirms. First Vertex AI integration, first trained artifact (linear probe), first real review gate with edits.

Est: 1.5 weeks. Depends on v4.1 (tissue mask, stain params, normalized patches).

1. Scope

In: 10× patch grid, Path Foundation embeddings (batched, cached), tumor-probability linear probe, heatmap + hotspot extraction, polygon editor UI, confirm gate, per-case cost logging.

Out: anything at 40×; probe fine-tuning on private data (probe v1 = public data only).

2. Patch grid + embeddings (`worker/triage.py`)

2.1 Grid

- Level selection: choose native level with MPP closest to 1.0 $\mu\text{m}/\text{px}$; resample patches to exactly 224 px @ 1.0 $\mu\text{m}/\text{px}$ (`read_region_srgb` + Macenko transform + `cv2.resize` if needed).
- Grid: stride 224 μm (no overlap), intersected with tissue mask (patch kept if $\geq 25\%$ tissue). Typical 2 cm^2 section $\rightarrow$ ~25–40k patches.
- Patch id: (`ix`, `iy`) grid indices; origin μm stored once in output header.

2.2 Path Foundation calls

- Vertex AI online prediction endpoint; request batches of 64 patches (base64 PNG or per endpoint spec), `asyncio` + `aiohttp`, concurrency semaphore 8, exponential backoff on 429/503 (max 5 retries).
- **Cache:** parquet at `artifacts/{slide_id}/embeddings/pathfoundation_{model_version}.parquet`, columns `ix`, `iy`, `emb` (`float32[384]`). Cache key includes model version + stain config hash. Re-runs and probe retrains are then free.
- Cost log: one `model_invocation` audit event per run: request count, wall time, estimated USD (`configs/pricing.yaml` unit price $\times$ calls).
- Runtime target: 30k patches $\leq$ 6 min wall (dominated by endpoint throughput; tune batch/concurrency against quota).

2.3 Tumor-probability probe (offline artifact, built during this stage)

- Training (one notebook, [research/probe_v1.ipynb](#), committed):
 - Data: **PatchCamelyon (PCam)** — 327k labeled 96 px patches from CAMELYON16 (public, downloadable) — upsampled to 224 @ ~1.0 $\mu\text{m}/\text{px}$; plus BACH 2018 (400 labeled ROIs) for 4-class sanity eval.
 - Push all patches through the same Path Foundation endpoint (~\$; cache).
 - Model: `sklearn.LogisticRegression(C=1.0, max_iter=2000)` on L2-normalized embeddings, binary tumor/normal.
 - Eval: held-out PCam test AUC (expect ≥ 0.95 per Path Foundation linear-probe reported range); calibration via Platt already inherent; record threshold sweep.
 - Artifact: [probe_v1.joblib](#) + [probe_v1.json](#) (training data hash, AUC, date, PF model version) → GCS [models/probe/](#); [weights_sha](#) into [model_versions](#) on every triage run.
- Known limitation (accepted, documented): PCam = metastatic lymph-node domain, not primary breast tumor. Probe v2 retrains on pathologist-corrected hotspots from the pilot (the review-edit flywheel) — deferred per decision #6.

2.4 Heatmap → hotspots ([pipeline/hotspots.py](#), pure)

```
def extract_hotspots(prob_grid: np.ndarray,      # [ny, nx] float32, NaN = no tissue

    grid_origin_um, stride_um,

    cfg) -> list[Hotspot]:

    # 1. gaussian_filter(prob_grid, sigma=cfg.sigma)      # default sigma=2 (~450  $\mu\text{m}$ )

    # 2. binary = smoothed >= cfg.prob_threshold        # default 0.5

    # 3. scipy.ndimage.label; drop components < cfg.min_area_mm2 # default 0.5 mm2

    # 4. score components by sum(prob); keep top cfg.max_hotspots # default 8

    # 5. polygon per component: marching squares (skimage.find_contours) on binary,

    #   simplify (shapely .simplify(tolerance=50  $\mu\text{m}$ )), dilate by cfg.margin_um (default 100)

    # 6. return polygons in  $\mu\text{m}$  + stats (area_mm2, prob_mean, prob_max)
```

All params in [configs/triage.yaml1](#). Unit-tested with synthetic gaussian blobs.

2.5 Output JSON

```
{ "heatmap_png_uri": "...",           // rendered at 1.25x, viridis, alpha in PNG

  "grid": { "origin_um": [0,0], "stride_um": 224, "nx": 400, "ny": 300},

  "prob_grid_uri": ".../probs.npy",

  "hotspots": [

    { "id": "hs_01", "polygon_um": [[x,y],...], "area_mm2": 3.2,

      "prob_mean": 0.81, "prob_max": 0.98, "source": "model", "excluded": false}],

  "model_versions": { "path_foundation": "...", "probe": "probe_v1@sha..." } }
```

3. Review UI

- Heatmap layer: single pre-rendered PNG stretched over slide bounds (OSD image overlay) with opacity slider — no per-tile heatmap tiles.
- Hotspot layer: polygons via **Annotorious OSD** (edit vertices, delete, draw new freehand/polygon). Each polygon tagged with stats tooltip.
- Pathologist actions → edit ops appended to a local list, POSTed on confirm:

```
[{"op": "modify", "id": "hs_01", "polygon_um": [...]},
```

```
{"op": "add", "id": "user_01", "polygon_um": [...]},
```

```
{"op": "exclude", "id": "hs_03", "reason": "DCIS only"},
```

```
{"op": "delete", "id": "hs_05"}]
```

- `POST /stages/triage/edits` stores diff; `POST /stages/triage/confirm` computes the **effective hotspot set** (machine output + diff), snapshots it into the confirm payload, queues `mitosis`.
- Rail shows count + total hotspot area; confirm disabled while zero hotspots (a case with no tumor → pathologist deletes all + explicit "no invasive tumor identified" checkbox → case routes to report stage with benign template; edge case must exist in v4.2 even if rare).

4. DB additions

CREATE TABLE hotspots (

id text NOT NULL, case_id uuid NOT NULL REFERENCES cases(id),

stage_execution_id uuid NOT NULL REFERENCES stage_executions(id),

polygon_um jsonb NOT NULL, area_mm2 double precision,

prob_mean double precision, prob_max double precision,

source text NOT NULL, -- model | pathologist_added | pathologist_modified

excluded boolean NOT NULL DEFAULT false, exclude_reason text,

PRIMARY KEY (case_id, id));

(Written at confirm time from the effective set — the DB holds confirmed truth; GCS holds machine raw.)

5. Tests

- `hotspots.py`: synthetic blob grids → expected component count, polygon area within 5%, margin applied; NaN handling (no-tissue) doesn't leak into smoothing (use `generic_filter` masking or fill 0 with weight mask).
- Embedding cache: second run makes **zero** endpoint calls (assert via injected fake client).
- Edit application: property test — `apply(diff, machine_output)` idempotent, order-stable.
- Probe eval numbers asserted in CI against the committed artifact metadata (guard against silently swapped weights).

6. Acceptance checklist

- ☐ Full run on a TCGA-BRCA slide: heatmap visually plausible (tumor regions hot, stroma/fat cold), ≤ 8 min wall.
- ☐ Vertex cost logged; visible in audit trail with USD estimate.
- ☐ Redraw a polygon, add one, exclude one, confirm → `hotspots` table matches effective set; `review_edits` diff stored; re-opening the page reproduces exactly the confirmed state.

- ☐ Recall spot-check on 5 archive slides with known tumor: pathologist-judged invasive area $\geq 95\%$ covered by proposals (informal now; formal in v4.6).
- ☐ Zero-tumor path works end to end (all deleted $\rightarrow$ benign flag $\rightarrow$ skips to report queue).